

Game Model (Prisoner's Dilemma)

Classic game where 2 players choose to cooperate or defect.

Each agent action: $a \in [0, 1]$ (cooperation degree).

Payoff: T =temptation, R =reward, P =punishment, S =sucker

$$R(a_i, a_j) = Ra_i a_j + T(1 - a_i)a_j + Sa_i(1 - a_j) + P(1 - a_i)(1 - a_j)$$

SARSA + Boltzmann Policy

SARSA: on-policy RL algorithm learning Q -values from (s, a, r, s', a') tuples.

Q-function: expected cumulative reward for action a in state s

$$Q(s_t, a_t) \leftarrow Q + \alpha(r_t + \gamma Q(s_{t+1}, a_{t+1}) - Q)$$

Boltzmann: soft action selection (β =inverse temperature)

$$\pi(a) = \frac{e^{\beta Q(a)}}{\sum_{a'} e^{\beta Q(a')}}$$

Stationary Distribution

γ : discount factor (future reward weight), α : learning rate

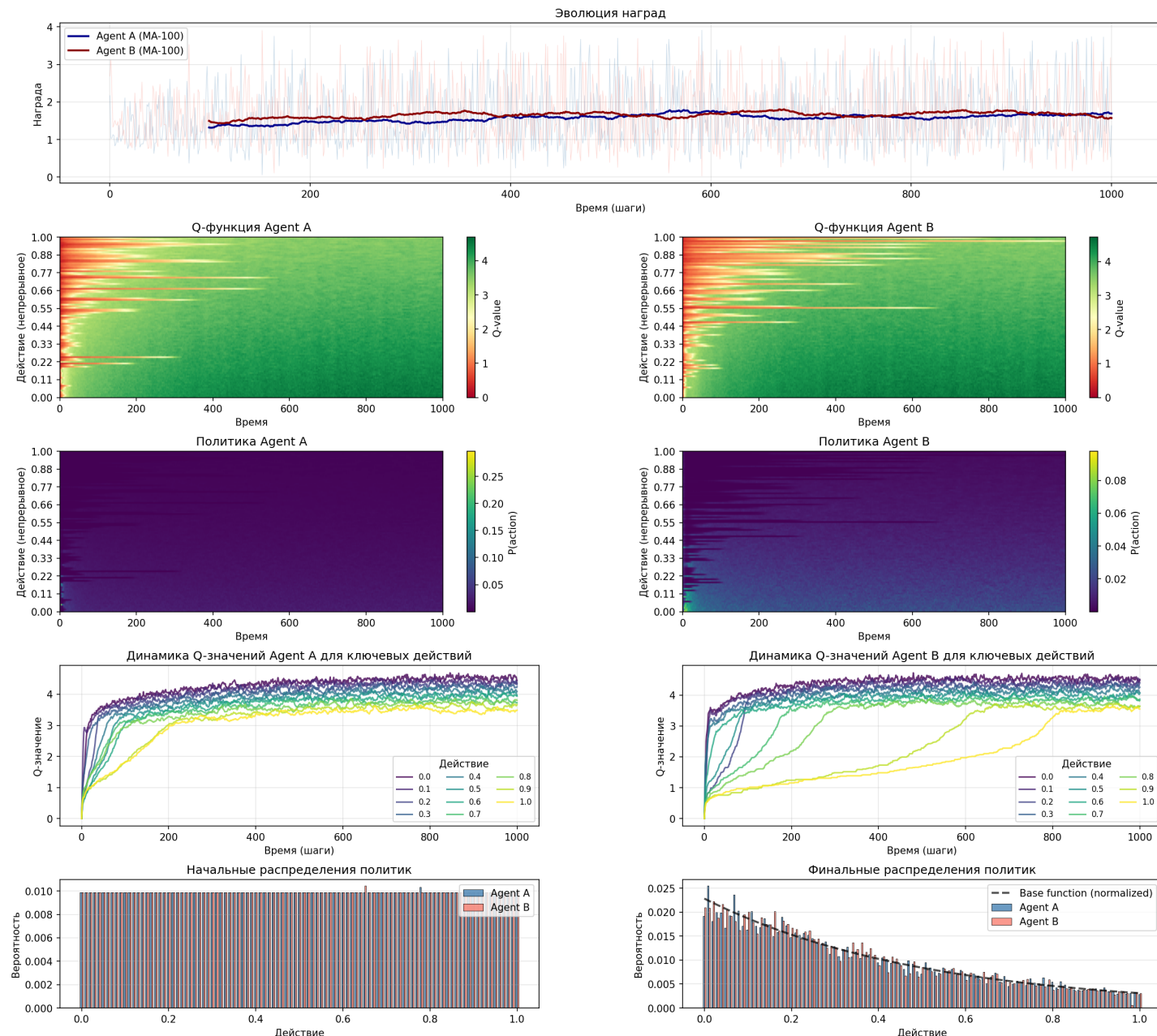
$$\text{At convergence: } \pi^*(a) = \frac{e^{\beta Q^*(a)}}{\sum_{a'} e^{\beta Q^*(a')}}$$

Bellman equation: $Q^*(a) = \mathbb{E}[r(a, a') + \gamma Q^*(a')]$

Key: $\pi^*(a)$ independent of γ , only convergence speed varies. Also we could say that about Q-learning with argmax.

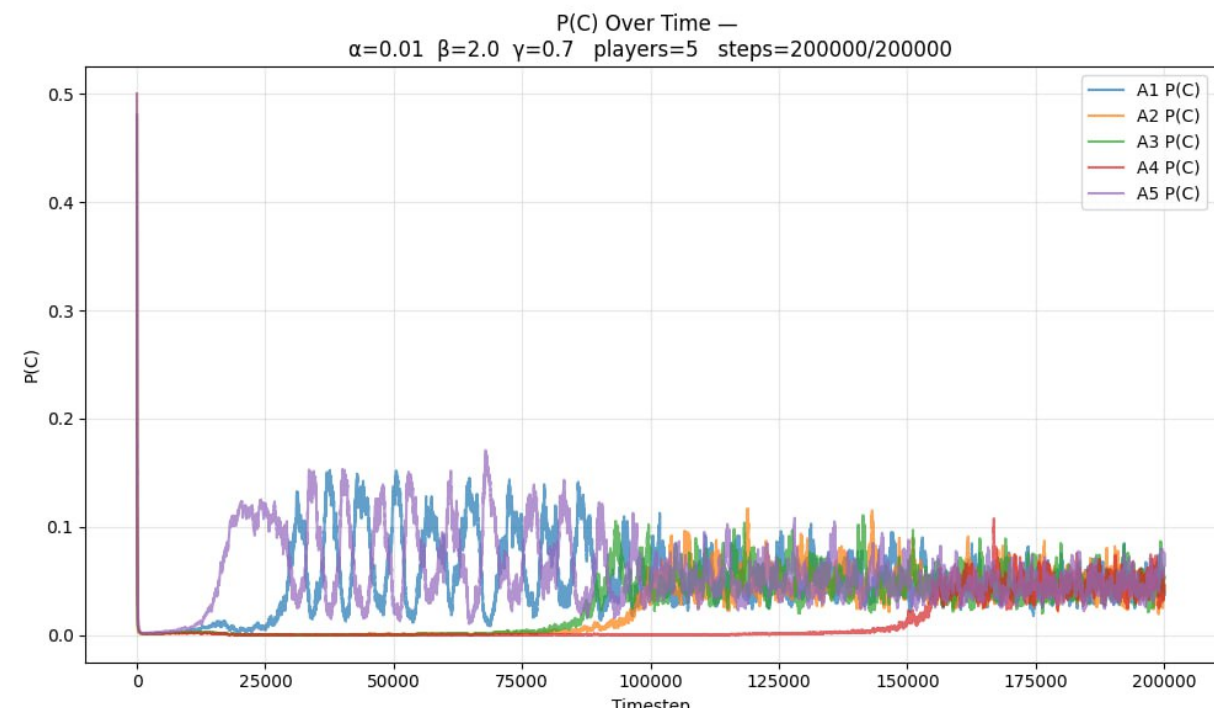
SARSA: Two-Player Results

Graphic for $R = 3, T = 5, P = 0, S = 1, \beta = 2.0, \gamma = 0.6$, steps = 2M



Q-Learning & Results: Multi-Player (N=5)

High β : escaping strategic trap causes oscillations.



Continuous & Multi-Player Extensions: Continuous action space $[0, 1]$ enables agents to learn nuanced mixed strategies rather than pure cooperate/defect. Multi-player games ($N=5$) exhibit complex phase-shifted oscillations between cooperation and defection states. Strategic trap: agents initially cooperate but eventually discover defection yields higher payoff, triggering collective shift. Oscillation mechanism: when all defect, payoffs drop; some agents rediscover cooperation, restarting the cycle.

Key Findings: Higher β (lower temperature) leads to sharper policy distributions and faster convergence toward defection equilibrium. Discount factor γ affects convergence speed but not the final stationary equilibrium distribution.

NLHF: Reward Model with KL Regularization

Train reward model from preferences with regularization to reference policy π_0 .

Bradley-Terry preference model: $P(y_w \succ y_l | x) = \sigma(r_\theta(x, y_w) - r_\theta(x, y_l))$

where σ is the sigmoid function, r_θ is the learned reward model

NLHF Loss with KL regularization:

$$\mathcal{L} = -\log \sigma(r_w - r_l + \tau \cdot (\log \pi_0(y_w | x) - \log \pi_0(y_l | x)))$$

where r_w, r_l = rewards for chosen/rejected, τ =KL weight

KL penalty prevents reward hacking: $D_{KL}(\pi || \pi_0) = \mathbb{E}_{y \sim \pi} \left[\log \frac{\pi(y|x)}{\pi_0(y|x)} \right]$

Regularization term anchors learned rewards to base model preferences, preventing over-optimization

NLHF: Implementation Pipeline

SFT: Supervised Fine-Tuning. **LoRA:** Low-Rank Adaptation.

Steps:

- SFT on TL;DR dataset (15k samples, 3 epochs)
- Create $N=5$ diverse policies via weight perturbation
- Generate 100 summaries/policy (batched, validated)
- Form preference pairs: heuristic scoring or human labels
- Compute reference log-probs $\log \pi_0(y)$ for KL term
- Train Reward Model (LoRA $r=8$) with NLHF loss

Model: Qwen2.5-3B-Instruct, LoRA, $\tau=0.1$, $lr=2e-5$

NLHF: Quality & Results

Generation quality controls:

- Multi-factor scoring: length, coherence, diversity
- Garbage detection: repetition, non-printable chars
- Validation with retry (2 attempts per prompt)
- Batched inference with gradient checkpointing

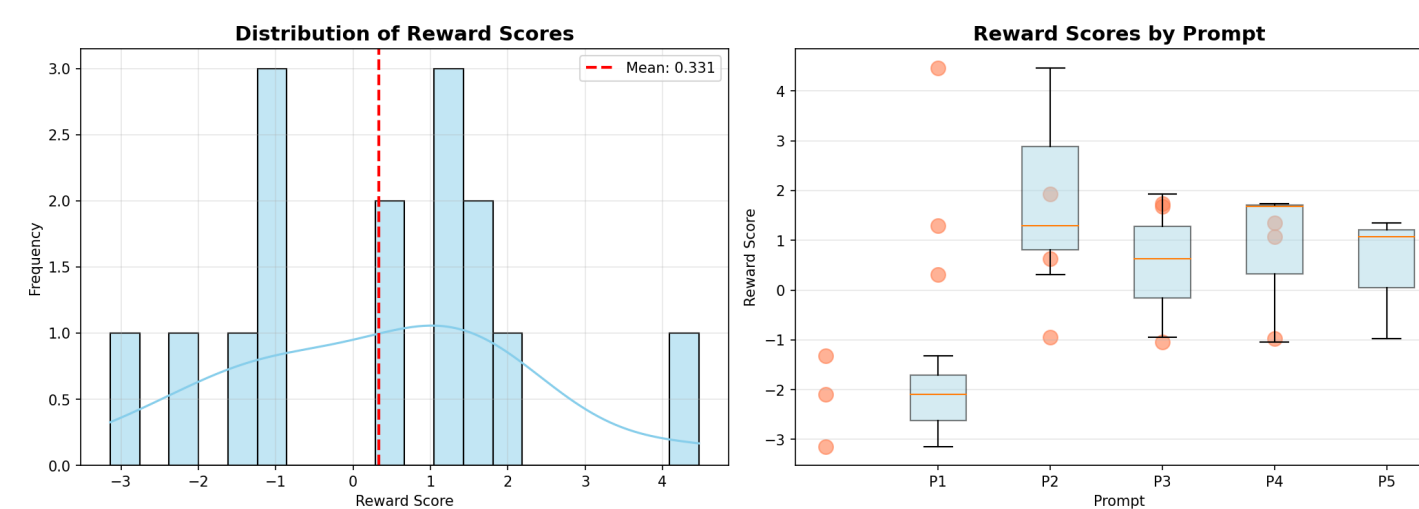
Data options:

- Synthetic pairs (heuristic): accuracy $\sim 60\%$
- OpenAI human feedback: accuracy $\sim 80\%$

Memory optimizations: Sequential forward passes, reduced batch size (2), gradient accumulation (32)

NLHF: Training Results

Reward Distribution Analysis: Mean reward = 0.331, diverse responses per prompt



Future Work: Nash-MD & Game Theory

Current: Reward Model training. **Next:** Multi-agent equilibrium learning.

Nash-MD (Mirror Descent): iteratively update policy toward Nash equilibrium

$$\pi_{t+1} = \arg \max_{\pi} [\eta P(\pi \succ \pi_t^\mu) - \text{KL}(\pi || \pi_t^\mu)]$$

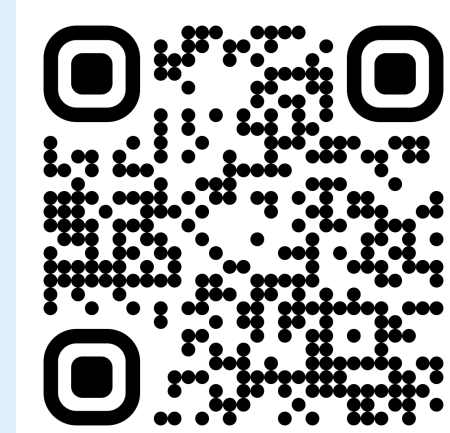
where $\pi_t^\mu = (1 - \eta\tau)\pi_t + \eta\tau\mu$ (mixture with reference μ)

Nash equilibrium: no player benefits from unilateral deviation

$$\mathbb{E}_{y \sim \pi^*} [P(y \succ y')] \geq \frac{1}{2}, \forall \pi'$$

Convergence guarantee: $O(\log T / \sqrt{T})$ to approximate Nash

Open questions: Multi-player dynamics, stability analysis



GitHub Repository